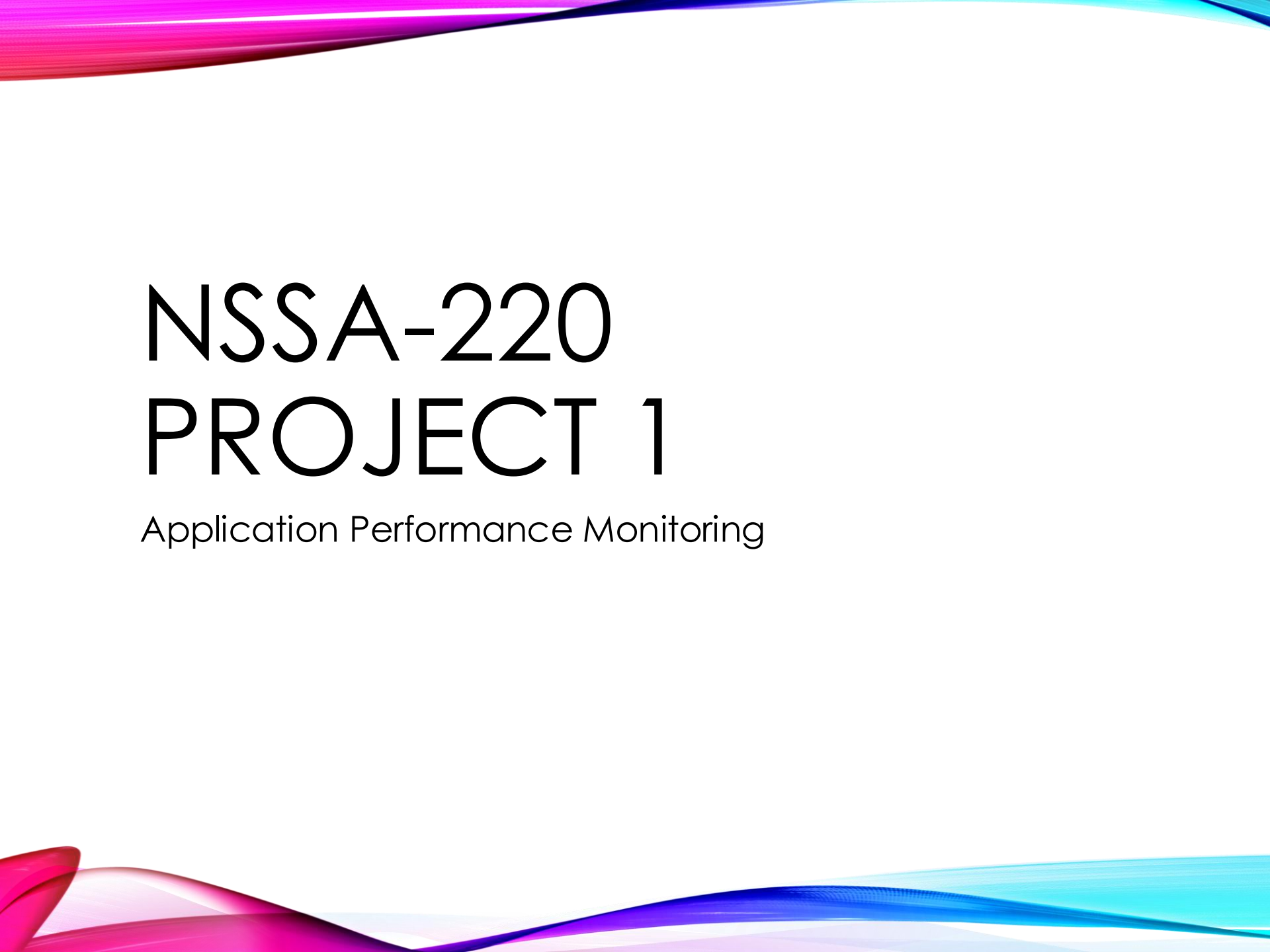




NSSA-220 PROJECT 1

Application Performance Monitoring

APPLICATION PERFORMANCE MONITORING (APM)

- Monitoring application performance is done for many reasons including, but not limited to:
 - Detecting systemic problems with software, such as memory leaks
 - Determining if computing resources are sufficient for the mix of software being used (number of CPUs, network bandwidth, memory, disk space, disk speed)
 - Detecting malicious activity in a system

POINTS OF INSPIRATION

- APM is highly relevant to just about any career path in computing, especially systems engineering, network management, network and system security, and system administration
- Employers are impressed by anyone that has experience with APM and analyzing APM data is highly related to the field of data analytics/data science

PROJECT 1 PRELIMINARIES

- Done in teams of 3
 - Declare your team on myCourses discussion area
 - Yes, you must do this in teams! No lone wolves allowed!
- Projects are designed to be more difficult than labs and require your own research beyond what you've been taught so far (worth 15% of your grade)

APM TOOL OVERVIEW

- You'll be writing an APM tool in Bash that will monitor a mix of processes in the form of executable programs written in C
- The script will start the processes, collect performance metrics, and perform a clean up at the end to kill these processes and any other processes that your script spawns

APM TOOL OVERVIEW (CONTINUED)

- The APM tool will monitor application performance by collecting metrics at both:
 - Process level
 - System level
- Your final submission will include a short report with Excel plots of the performance metrics collected

PROCESS LEVEL METRICS

- Linux provides the ability to collect performance metrics at the process level for both CPU and memory
- Using `ps`, you can see the %CPU and %memory that any process is using
- If the applications are utilizing most of or all of the CPU, you may consider adding cores to a VM or processors to a physical machine
- Unbounded memory growth over time indicates poor memory management

SYSTEM LEVEL METRICS

- Disclaimer: the following metrics could be measured at a process level, but not without great pain
- You will monitor these system level metrics
 - Network bandwidth utilization
 - Hard disk access rates
 - Hard disk utilization

NETWORK BANDWIDTH UTILIZATION

- Network bandwidth utilization indicates how much bandwidth each network interface card is using over time
- You will measure bandwidth utilization in terms of receive (RX) data rate and transmit (TX) data rate using the ifstat tool
- If data rates are approaching network capacity, additional capacity and/or load balancing is likely needed

HARD DISK ACCESS RATES

- Hard disk access rates are typically measured in volume of reads and writes per second
- Monitoring disk access activity over time may show that a solid state drive is needed over a mechanical drive or alert a system administrator to unexpected excessive access rates
- You will measure hard disk writes in kB/second to the primary hard drive (sda) using the iostat tool

HARD DISK UTILIZATION

- Monitoring hard disk utilization, in terms of remaining disk capacity, is not as critical as it once was, but there are still times where unexpected reductions or increases in disk capacity may occur
- You will measure hard disk utilization on the "/" mount (the /dev/mapper/centos-root filesystem) using the df tool

CENTOS VM RESOURCES

- The default CentOS VM resources are
 - 1 CPU core
 - 2 GB memory
 - 100 GB hard disk
 - CentOS NIC (ens33) in NAT mode
- You are required to use the default CentOS VM resources to collect data for this project – *MAYBE...*

APPLICATIONS TO MONITOR

- You will be given 6 applications in the form of pre-compiled C executables
- Each application will take a single argument: the IP address of a VMNet NIC on the lab PCs or any virtual NIC on your own computer
- You will need to run your VM in “NAT Mode” to allow connectivity between the VM and the host’s virtual NIC. The virtual NIC is initially disabled.

APM TOOL REQUIREMENTS

1. The APM tool shall collect process and system level metrics every 5 seconds
2. The APM tool shall collect %CPU and %memory utilization per process using the ps tool
3. The APM tool shall collect network bandwidth utilization in terms of RX data rate and TX data rate (kB/s) with a sampling interval of 1 second on the ens33 interface using the ifstat tool

MORE APM TOOL REQUIREMENTS

4. The APM tool shall collect hard disk writes in kB/second to the primary hard drive (sda) using the iostat tool
5. The APM tool shall collect hard disk utilization of the "/" mount in Megabytes available using the df tool
6. The APM tool shall output all CPU and memory metrics to a CSV file specific to the process they were measured from. Name the files <proc_name>_metrics.csv

MORE APM TOOL REQUIREMENTS

7. The format of the process specific output files shall be
 - <seconds>, <%CPU>, <%memory>
8. The APM tool shall write all system level metrics to a file called *system_metrics.csv*
9. The format of the system level output file shall be
 - <seconds>, <RX data rate>, <TX data rate>, <disk writes>, <available disk capacity>

MORE APM TOOL REQUIREMENTS

- 10. The APM tool shall spawn all application processes
- 11. The APM tool shall kill all application processes and any other processes it spawns in an exit trap function called “cleanup”
- 12. The APM tool shall minimally include functions to (1) spawn applications and other processes, (2) collect process level metrics, and (3) collect system level metrics

MORE APM TOOL REQUIREMENTS

13. Your team's submission to myCourses shall be a single zip file including the APM tool script, all output files for a 15 minute run, and a report showing Excel plots of the metrics collected. The report format is provided.
14. Your application performance data shall be measured using the default VMWare resources

GRADING

- See grading sheet for details
- Major point loss will occur if an individual exhibits poor teamwork. You will fill out a peer review form to indicate how well the team worked together. Let me know ASAP if there are issues with your team.

HINTS

- Read the man pages for the specified tools!
- One of the tools requires you to spawn an additional process to use it.
- All the values in the output files should be numbers only. You already know what the units are.
- Put all the requirements into a table in a Word document and check them off as you complete them.

ASK FOR HELP!

- Don't suffer in silence. Ask me or your TA for help sooner rather than later!
 - Attend my office hours or the TA's
 - Make an appointment outside of office hours
 - Send an email
- If you're not sure if you've met a specific requirement, please ask!